SCHOOL OF BUSINESS

COURSE 1: INTRODUCTION TO LARAVEL

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2025 – 2026

COURSE OUTCOMES

- ✓Match each given URL to a specific process using **routing**
- ✓Group related processes using **controllers**
- ✓Retrieve data from an HTTP query using Request
- ✓Return content in text, HTML, JSON, etc. formats using Response
- ✓Integrate data into templates using Blade
- ✓Interact with the user using forms
- ✓Create, update, and track changes to a database system using migrations
- ✓Facilitate communication with a database using Eloquent

SUMMARY

PRELIMINARIES

1. What is a framework?
2. Why Laravel
3. Prerequisites for installing Laravel

WHAT IS A FRAMEWORK?

- Problem: How to efficiently develop an application (speed, best practices, etc.)
- Solution: Set of "Ready-to-use" components and recommendations = Framework

WHAT IS A FRAMEWORK?

Benefits of a framework:

- Code structure (MVC model)
- Database abstraction
- Reuse of tried and tested components (email, users, etc.)
- Implementation of good coding practices
- Ease of code maintenance and evolution
- Facilitates teamwork
- Strong community (support and updates)

WHAT IS LARAVEL?

- Laravel is one of the most widely used PHP frameworks.
- Created by Taylor Otwell in June 2011.
- Latest major release: 11.0 (March 2024).
- Giuthub's highest-rated PHP framework in 2016.
- Laravel is still based on Symphony for at least 30% of its frameworks.
- Competition with Symphony, CodeIgniter, and ZendFramework.

WHAT IS LARAVEL?

- **Free and open source:** Laravel is free and open source, and its community actively contributes to its development.
- **MVC approach:** Laravel uses the Model-View-Controller (MVC) pattern to organize application code.
- **Simplicity and ease of learning:** Laravel is designed to be accessible to beginners while offering powerful features for experienced developers.

Taylor Otwell created Laravel in response to the complexity of CodeIgniter, aiming to offer a more flexible and easier-to-learn alternative.

LARAVEL: MAIN FEATURES

- Laravel stands out for several key features:
- **Modern PHP:** Laravel leverages the latest PHP features.
- **Fast Build**: Setting up a Laravel project is quick and easy.
- **Object-Oriented Development**: Laravel promotes object-oriented programming best practices.
- **Centralized Dependency Management**: Laravel uses Composer, a dependency manager for PHP.

MVC ARCHITECTURE

It's a code organization model:

✓The Model is responsible for managing the data

✓The View is responsible for formatting for the user

✓The Controller is responsible for managing everything

➔ Generally, we summarize by saying that the Model manages the database, the View produces the HTML pages, and the Controller does everything else.

MVC ARCHITECTURE

In Laravel:

- ✓**model** corresponds to a table in a database. It's a class that inherits from the Model class and allows for simple and efficient management of data manipulation and the automated establishment of relationships between tables. Models are located in App/Models/.
- ✓**Controllers** fall into two categories: classic controller and resource controller. They are located in App/Http/Controllers/.
- ✓**Views** are either a simple file with HTML code or a file using Laravel's Blade templating system. They are located in resources/views/.

MVC ARCHITECTURE

In summary, Laravel's MVC architecture consists of three main parts:

✓ **Model**: Manages data logic and database interactions via the Eloquent library (ORM: Object-Relational Mapping).

✓ **View**: Manages data presentation, primarily through the Blade templating engine.

✓ **Controller**: Manages application logic and acts as an intermediary between the Model and the View.

HOW LARAVEL WORKS

This figure represents the key components of a Laravel application:
DATABASE: Stores application data.
MODEL: Manages data and business logic, interacts with the database.
ROUTES: Directs HTTP requests to the appropriate controllers.
CONTROLLER: Processes requests, interacts with the model, and returns data to the view.
VIEW: Displays data to the user via the user interface.
This figure shows how Laravel structures an application by separating responsibilities between these components for efficient data and query management.